1. Chemical Reactions

1.1. Physical and Chemical Changes

Physical Change	Chemical Change
The reaction is readily reversible	The reaction is harder to reverse
The product has no new chemical properties	Chemical product has different properties
Ex. dissolving a solute in a solvent	Energy change [exothermic/endothermic]

1.2. Rates of Reaction

Collision Theory

Successful collisions have enough activation energy to break pre-existing bonds and form new bonds at impact.

Rates of reaction

- The measure of the speed of the collision
- Calculated by the concentration of reactant used up or product produced per unit of time
- Unit = $(mol/dm^3)/s$

1.3. Concentration

Increasing the concentration of reactants increases the rate of reaction

- Higher-concentration reactants contain more particles per unit volume, increasing the successful collision and reaction rates.
- Increasing concentration increases the number of particles per unit volume, leading to more frequent collisions and a faster reaction rate (activation energy unchanged).

Less particles, less frequent and successful collision

More particles, more frequent and successful collision

1.4. Temperature

Increasing temperature increases the **rate of reaction**

- Increasing temperature gives particles more kinetic energy. A greater proportion of reactant particles have energy ≥ activation energy, leading to more frequent successful collisions and a faster reaction rate.
- In a reversible reaction, increase in temperature increases the rate of reaction for both the forward and backward reaction.

Particles have less energy, less frequent and successful collision

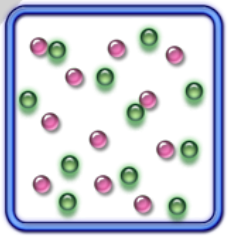
Particles have high energy, more frequent and successful collision

1.5. Surface Area of Solids

Decreasing the particle size increases the rate of reaction

- Decreasing particle size increases surface area; more reactant particles are exposed to collide, so the successful collision rate increases. This results in an increased rate of reaction

Less particles, less frequent and successful collision



More particles, more frequent and successful collision

Explosive Combustion

- Fine particles are combustible in the air due to a larger surface area
- The rate of reaction is high, making them explosive
- Examples: methane gas in coal mines and flour milling

1.6. Pressure

Increasing the pressure in a gaseous system increases the rate of reaction

- The distance between particles is reduced under pressure
- There are more particles per unit volume; the successful collision rate increases, resulting in an increased reaction rate.

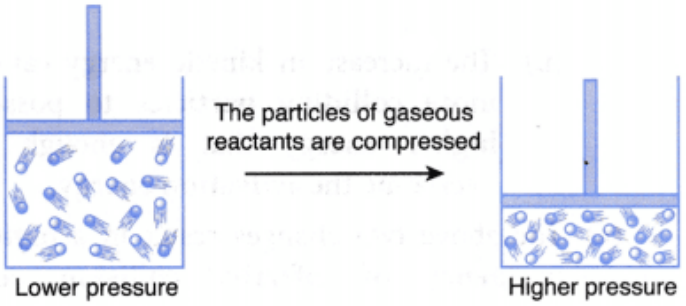


Figure *Increase in pressure of gaseous reactants*

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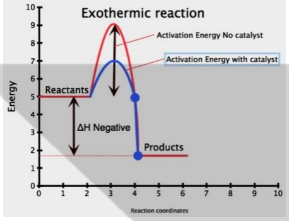
1.7. Catalyst

A catalyst is a substance that increases the reaction rate **by lowering the activation energy and is left unchanged at the end of the reaction.**

- More particles will have an energy greater than or equal to the activation energy, therefore successful collision rate increases, resulting in an increased rate of reaction
- For gaseous reactants, if the catalyst is solid metal, the catalyst provides a surface for the reaction.
- The larger the surface area of the metal catalyst, the larger the area for the reaction to take place; therefore higher the rate of reaction



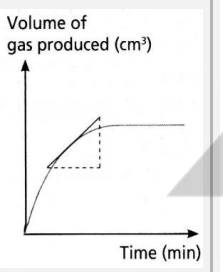
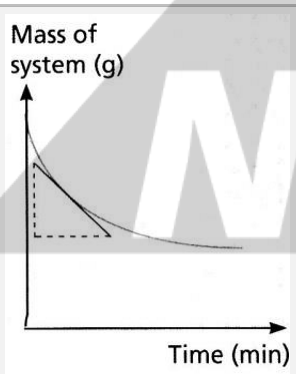
Enzymes Affecting Rate of Reaction

- Enzymes are biological catalysts that speed up reactions but remain chemically unchanged.
- Enzymes function best at optimum temperature and pH level; otherwise, they may denature and completely stop functioning.

1.8. Rates of Reaction Graphs

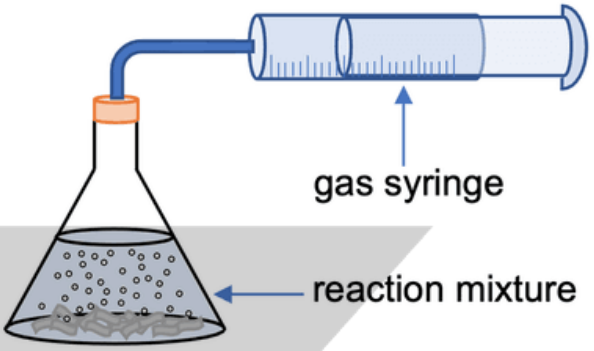
Interpreting graphs:

A graph with a steeper gradient at the beginning and reaching a horizontal gradient faster depicts a high rate of reaction.

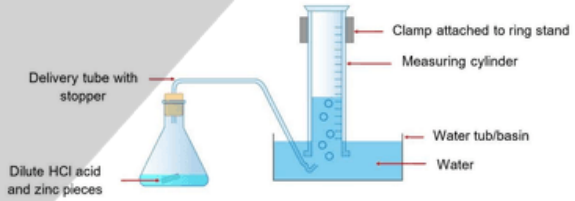
Gas Evolved	Mass Loss	Colour Change
		
If a gas evolves, measure the volume of gas produced per unit of time using a gas syringe	If a gas evolves, measure the loss in mass per unit time by placing it on a balance and then putting cotton wool on top to allow gas to pass but not enter	If a colour change, measure the time taken to turn cloudy

1.9. Evaluating Practical Methods

1. **Change in Mass method**
   - Not suitable for experiments where hydrogen gas is given off (because density decreases, too small mass)
2. Easier to set up the **gas syringe** and obtain reliable measurements



3. **An inverted Measuring cylinder** to collect gas over water is harder to set up.



1.10. Reversible Reactions and Equilibrium

The symbols of reversible reactions are: ⇌

- The products can then react with each other or decompose to form the reactant molecules.
- Two equations you should know:
- $CuSO_4 \cdot 5H_2O$ (blue) $\rightleftharpoons CuSO_4$ (white) $+ 5H_2O$
- $CoCl_2 \cdot 6H_2O$ (pink) $\rightleftharpoons CoCl_2$ (blue) $+ 6H_2O$

(anhydrous by heating; hydrated form by adding water)

Reversible Reaction in a closed system when equilibrium:

- Rate of forward reaction = rate of reverse reaction
- Concentrations of all reactants and products remain constant and are no longer changing

Equilibrium

- **Le Châtelier's Principle:** if conditions of equilibrium are changed, the position of the equilibrium moves to oppose change
- **Temperature:** Increase in temperature shifts the equilibrium in the endothermic direction. When the temperature is lowered, equilibrium shifts in the exothermic direction.
- **Pressure:** Pressure raised; equilibrium moves to the side with the fewest gas molecules. When pressure is lowered, equilibrium shifts to the side with most gas molecules.

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- **Concentration:** Decreasing reactant concentration or increasing product concentration; equilibrium moves to the reactant side. Increasing reactant concentration or decreasing product concentration; equilibrium shifts to the product side.
- **Catalyst:** The presence of a catalyst does **not** affect the equilibrium position but increases the rate at which equilibrium is reached. Therefore, it only increases the rate of the forward and backward reactions by the same amount.

Oxidising Agents are **Electron Acceptors** and Reducing Agents are **Electron Donor.**

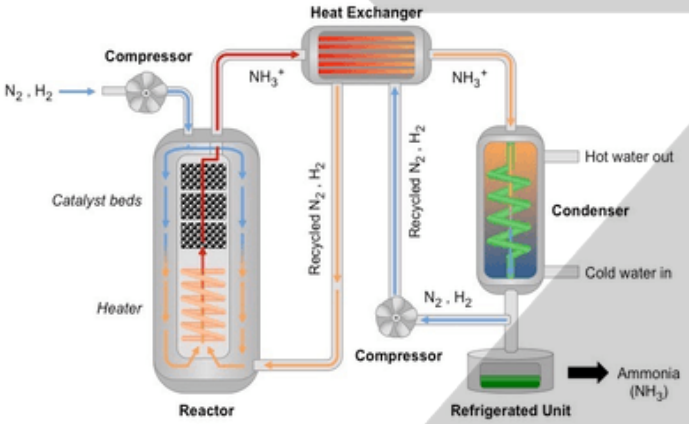
1.11. Haber Process

$N_2 + 3H_2 \rightleftharpoons 2NH_3$ (g) [Exothermic]

Materials (Reactants) in the Haber Process

Nitrogen: From the **air.**

Hydrogen: From the reaction between **Methane and Steam**



Conditions

Temperature: **450°C**

Pressure: **200 atm (20,000 kPa)**

Catalyst: **Iron catalyst**

Why are these conditions chosen in the Haber Process?

1. **Pressure is not extremely high** - (Although increased pressure is more yield, more pressure will require more expensive equipment, and it can explode).
2. **Optimum temperature** - (If lower temperature, more ammonia yield, however too low, the rate is uneconomical.

1.12. Contact Process

$2SO_2 + O_2 \rightleftharpoons 2SO_3$ (g) [Exothermic]

Materials (Reactants) in Contact Process

Sulfur Dioxide: **Burning sulfur or Roasting Sulfide ores**

Oxygen: From the air

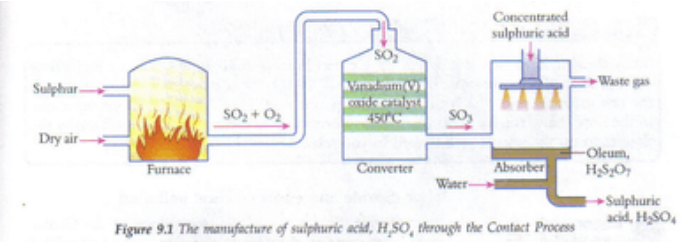


Figure 9.1 The manufacture of sulphuric acid, H_2SO_4 through the Contact Process

Conditions

Temperature: **450°C**

Pressure: **2 atm (200 kPa)**

Catalyst: **Vanadium (V) oxide catalyst**

Why are these conditions chosen in the Contact Process?

1. **Pressure is high** - (Increase pressure, more yield)
2. **Optimum temperature** - (If lower temperature, more sulfuric acid yield, however too low, the rate is uneconomical.

1.13. Redox

Redox: A simultaneous oxidation and reduction reaction

Oxidation	Reduction
Loss of electrons	Gain of electrons
Gain of oxygen	Loss of oxygen
Loss of hydrogen	Gain of hydrogen

NOTE: **Roman Numerals** next to an element's name are the **oxidation number of an element in a compound**, e.g. Iron (II) and Iron (III). The reaction product formula depends on this.

Reducing agents are oxidised, and oxidising agents are reduced.

Identifying Redox Reactions

1. **Oxidation State:**

The oxidation state highlights electron movement in a reaction

eg. $CuO + Mg \rightarrow MgO + Cu$

$Cu^{2+} + Mg \rightarrow Mg^{2+} + Cu$ [oxide is a spectator and is removed as it doesn't change its oxidation state]

Here Copper (II) is reduced while magnesium is oxidized; the reaction is redox

2. **Indicators:**
   - **Acidified Aqueous Potassium manganate (VII)** is a deep purple oxidising agent; when added to a reducing agent changes from purple to colourless.
   - **Aqueous Potassium iodide** is a reducing agent; when added to an oxidising agent, it changes from colourless to yellow-brown.

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Redox Reactions by Changes in Oxidation Number

Rules	Example
The oxidation numbers of the element in their uncombined state are zero	In H_2, the oxidation number of H is "0."
The oxidation number of a monatomic ion is the same as the charge of the ion.	In Zn^{2+}, the oxidation number is "+2."
The sum of the oxidation numbers in a compound is zero	In H_2O, the oxidation number of H is +1, and O is -2, (2 x (+1)) + (1 x (-2)) = 0
The sum of the oxidation numbers in an ion is equal to the charge of the ion	In MnO_4-, the oxidation number of Mn is +7, therefore (1 x (+7)) + (4 x (-2)) = -1

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